



SPECIAL ISSUE ARTICLE

A Preliminary Investigation of Associations Between Traumatic Events Experienced During Pregnancy and Salivary Diurnal Cortisol Levels of Brazilian Adolescent Mothers and Infants

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ABSTRACT

Introduction: Interpersonal violence against women is a major global health problem that may have intergenerational effects. This study investigated associations between maternal experiences of interpersonal violence and other traumatic events and maternal and infant salivary diurnal cortisol in a cohort of adolescent mothers in São Paulo, Brazil.

Method: Adolescent mothers (14–19 years) participating in a home-visiting intervention were interviewed retrospectively about lifetime and pregnancy violence and trauma exposure. Mothers collected saliva at waking and before bedtime from themselves ($n = 23$) and their infants ($n = 32$) at 12 months postpartum. Multivariable regression models were used to examine associations between trauma history variables and salivary diurnal cortisol.

Results: Adjusting for the intervention group, infant sex, maternal age, non-supplement medication use, and sample collection time, we found that higher-than-average lifetime trauma exposure was associated with maternal evening cortisol ($b = 0.472$, p -value = 0.028). Lifetime assaultive violence exposure was also associated with maternal evening cortisol ($b = 0.196$, p -value = 0.02). Maternal exposure to traumatic events in pregnancy was positively associated with bedtime cortisol levels of infants ($b = 0.21$, $p = 0.01$). Trauma variables were not associated with maternal or infant morning cortisol levels.

Conclusion: Results suggest that maternal trauma history influences both maternal and infant postnatal cortisol regulation as indexed by evening cortisol levels. These results are consistent with models of fetal programming; however, future studies should investigate potential postnatal psychobiological pathways. Lifetime trauma exposure may also become embedded in the

1 | Introduction

Interpersonal violence is a public health problem of global importance with implications for physical and mental health across the life course (Krug et al. 2002; Panter-Brick 2010; Ribeiro et al. 2009). Violence against women, in particular, is an issue for countries across the economic development spectrum (Garcia-Moreno et al. 2006). Pregnancy may be a particularly vulnerable period as pregnant women have a high prevalence of experiencing interpersonal violence (Campbell, García-Moreno, and Sharps 2004; Taillieu and Brownridge 2010). Violence experienced during pregnancy has consequences not only on women's health but may also have intergenerational effects, impacting infant health and physiology. Infancy, or the first 12 months of life, is a sensitive period of development in which developmental trajectories may be primed by prenatal exposures or shift in response to postnatal experiences (Lupien et al. 2009).

Prenatal exposure to violence has been associated with adverse birth outcomes and pregnancy health, as well as offspring health and development across the life course. Associated adverse birth outcomes and pregnancy complications include maternal mortality, miscarriage, preterm birth, low birth weight, and intrauterine growth restriction (Boy and Salihu 2004; Ferraro et al. 2017; Ferri et al. 2007; Hill et al. 2016; Kaye et al. 2006; Shah and Shah 2010). Such prenatal exposures, including both intimate partner violence (IPV) and other forms of trauma, have been associated with offspring emotion dysregulation, internalizing and externalizing disorder symptoms, and increased asthma incidence in infants and children (Bergman et al. 2007; Martinez-Torteya et al. 2017; Sancho-Rossignol et al. 2018; Suglia et al. 2009). These associations may be partially explained by the concept of fetal programming, which suggests that fetal development is sensitive to environmental cues and alterations in development may shape postnatal health and disease risk (Barker 1998). Environmental signals may be relayed to the developing fetus via shifts in maternal gestational physiology (Moisiadis and Matthews 2014).

Changes in maternal hormones such as cortisol may be one mechanism given its role in the physiological and psychological responses to stress. Cortisol is a hormone produced by adrenal glands as part of the hypothalamic–pituitary–adrenal axis (HPAA) in response to stressful or threatening stimuli. Cortisol has normative functions in pregnancy and fetal development, and levels, measured in blood, typically increase over pregnancy, peaking in the weeks before birth and rapidly declining immediately after birth and returning to pre-pregnancy levels within 2–12 weeks (Demey-Ponsart et al. 1982; Jung et al. 2011; Mastorakos and Ilias 2000). However, exposure to violence and trauma may disrupt these processes during pregnancy or in the postpartum.

Violence and trauma exposure have been shown to impact maternal HPAA dynamics as indexed by cortisol levels during

pregnancy. IPV exposure has been associated with evening salivary cortisol levels in pregnant Nicaraguan women between 18 and 39 weeks of gestation (Valladares et al. 2009). However, a study of Peruvian women did not find an association between IPV and hair cortisol levels in mid-pregnancy (Kelsall et al. 2023). A study of IPV in U.S. women did not find an independent effect of IPV exposure on salivary cortisol response to a stress task at 15–18 weeks of gestation but did find a significant interaction with depressive symptoms (Levendosky et al. 2024). Studies of prenatal trauma and maternal cortisol in pregnancy are more geographically diverse than studies of intergenerational impacts on infant cortisol. A racially and ethnically diverse study of pregnant women in the United States (U.S.) found a positive association between lifetime trauma exposure and average hair cortisol levels across pregnancy, specifically among Black women in the cohort (Schreier et al. 2016). However, another study found that trauma exposure was inversely associated with hair cortisol in a study of women living in Greece, Spain, and Perú and Peruvian women had the greatest trauma exposure and highest hair cortisol levels (Dobernecker et al. 2023). The variation in study results may be due to how exposure variables were operationalized, gestational age at sample collection, and sample type for cortisol analyses (i.e., hair vs. saliva).

Symptoms of posttraumatic stress disorder (PTSD) may further modify the associations between trauma and cortisol. The literature on PTSD symptoms and pregnancy cortisol levels is small, but several studies have reported significant associations. General PTSD symptoms have been negatively associated with the salivary cortisol awakening response at 12, but not 32, weeks of gestation (Bublitz et al. 2023), while women with dissociative PTSD have been shown to have higher salivary cortisol levels and the flattest diurnal curves from early pregnancy up to 6 weeks postpartum (Seng et al. 2018). Two recent reviews found that experiencing early life adversity was associated with an elevated cortisol awakening response, blunted cortisol response to stress, and alterations in diurnal cortisol rhythms across pregnancy (Epstein et al. 2021; Rinne et al. 2023). Notably, trauma exposure is associated with elevated cortisol levels even in the absence of clinically significant PTSD symptoms (Meewisse et al. 2007).

The impact of prenatal exposures on maternal cortisol after birth is less frequently studied than infant outcomes and has focused on the early postpartum period. Experiences of trauma may impact the normative decline of cortisol to pre-pregnancy levels. Childhood adversity is associated with a steeper decline in cortisol levels after a psychological lab stress test at 6 months postpartum (Brand et al. 2010). This same study found that mothers who reported an additional risk factor, including recent stressful life events or comorbid PTSD, had elevated cortisol in response to the stressor compared to unexposed women (Brand et al. 2010). A similar pattern has been observed for the cortisol awakening response. Women who reported experiencing early life adversity had a larger awakening response than those who did not experience

adversity at 2–6 months postpartum (Gonzalez et al. 2009). A longitudinal study investigating diurnal salivary cortisol levels from 16 weeks of pregnancy to 6 weeks postpartum found that women with a dissociative subtype of PTSD after trauma exposure had higher bedtime cortisol levels and a flatter diurnal curve (Seng et al. 2018). This study contributes to this literature by investigating associations with maternal postnatal salivary cortisol measures at 12 months postpartum.

Some prior studies have documented intergenerational impacts of maternal trauma exposure, with specific foci on IPV and PTSD symptoms. Several studies suggest that maternal IPV victimization in the prenatal period is associated with differences in salivary cortisol reactivity to stressors in infancy and childhood (Cordero et al. 2017; Levendosky et al. 2016; Martinez-Torteya et al. 2016; Slopen et al. 2018). Studies also suggest that witnessing IPV in childhood or experiencing other forms of early life adversity may also produce similar intergenerational consequences for offspring cortisol (Brand et al. 2010; Broeks et al. 2023; Khoury et al. 2021; Parade et al. 2019). Maternal preconception and prenatal PTSD symptoms may also be related to salivary cortisol levels in infants and children (Guardino et al. 2022; Yehuda et al. 2005). Investigations into other forms of trauma and violence exposure are limited in number. A study on other trauma types found an association between maternal prenatal trauma exposure and hair cortisol levels in their 3- to 4-year-old children (Slopen et al. 2018). Much of this prior work has largely been conducted with populations in the United States and Europe.

1.1 | Context for This Study

A limitation of the existing prenatal stress and trauma literature is a consistent focus on ethnically homogeneous, clinical, and relatively high-income sample populations in the United States and other high-income countries (Bush et al. 2017; Glover et al. 2018; Thayer and Kuzawa 2014). This gap is limiting as exposure to traumatic events, and other forms of psychosocial stress, is likely to differ dramatically across income lines, within and between countries, and result in more frequent and severe exposure for low-income and ethnic minority groups (Lantz et al. 2005; Menard, Bandeen-Roche, and Chilcoat 2004). As others have noted, a “focus on high-risk populations will address the challenges and complexities in generalizing extant stress research findings to these communities and the importance of understanding the impact of adversity within populations more chronically exposed to severe stressors” (Bush et al. 2017, 1554). This preliminary study expands on this literature by investigating associations between maternal trauma and violence exposure and stress-related changes in maternal and infant salivary cortisol production at 12 months postnatal in a population of adolescent mothers and infants living in São Paulo, Brazil. Furthermore, many studies have focused exclusively on IPV exposure. While such exposures are immensely harmful to maternal and child health, mothers may also experience other forms of traumatic events. This study aimed to address some of these limitations by examining the programming effect of trauma and violence exposure using a locally developed measure in a high-risk population of first-time adolescent mothers and their 12-month-old infants.

In this study, we conducted work with pregnant adolescents living in low-income areas of São Paulo, Brazil. Adolescent pregnancy is a common public health issue in low- and middle-income countries, and Brazil is above the global and Latin American average rates (Lancet 2020). Poverty is a risk factor for adolescent pregnancy, and women who were pregnant during adolescence report lower educational attainment and income later in life (Brahmbhatt et al. 2014; Cone et al. 2021; Ganchimeg et al. 2014). Experiencing trauma and violence are risk factors for adolescent pregnancy, and adolescent mothers often experience higher rates of IPV than non-pregnant adolescents and older mothers (Bott et al. 2019; Devries et al. 2010; Shadigian and Bauer 2004). The experience of violence or trauma before or during pregnancy may contribute to the elevated rates of negative birth outcomes, postpartum depression, anxiety, and PTSD among adolescent mothers (Ferri et al. 2007; Thomas et al. 2019). Such factors may contribute to an intergenerational cycle of poverty and stress.

Adolescence is a dynamic life stage during which individuals navigate multiple energetic demands and trade-offs in investments in growth and reproductive development. The unique social, developmental, and hormonal dimensions of and trade-offs occurring during adolescent pregnancy may exert distinct influences on the intergenerational transmission of stress. The associations between stress and biomarkers such as cortisol may become decoupled during adolescent pregnancy as cortisol plays normative roles in energetic trades occurring in adolescence and mediates a functional crosstalk between the HPA and the hypothalamic–pituitary–gonadal axis (Marceau et al. 2015; Viau 2002). Several studies report no or inverse associations between depression, anxiety, or general psychological stress and cortisol sampled from pregnant adolescents (Dorn, Susman, and Petersen 1993; Rondó et al. 2004; Spicer et al. 2013). However, these and other studies found associations between adolescent maternal mental health and biomarkers with adverse birth and infant outcomes (Ponirakis, Susman, and Stifter 1998; Spicer et al. 2013), suggesting that stress is still transmitted to the developing offspring. The impacts on the postpartum HPA function after adolescent pregnancy are even less well understood.

Using data collected from a cohort of adolescent mothers and their infants, recruited from the western regions of São Paulo, Brazil, and participating in the *Primeiros Laços* home visit intervention, we conducted preliminary analyses to test for associations between exposure to traumatic events and maternal and infant HPA function at 12 months of age. These regions are characterized by high rates of adverse living conditions, poverty, and urban violence (Ferri et al. 2007; Jacobi 1994). The adolescent birth rate is high in this context, and Brazilian adolescent mothers often live in impoverished conditions, facing high rates of exposure to violence and mental health problems (Chalem et al. 2012; Ferri et al. 2007; Pinheiro et al. 2012).

1.2 | Research Questions

In this study, we examined retrospective reports of lifetime exposure to violence and other traumatic events, as well as exposure during pregnancy, and associations with maternal and infant

diurnal salivary cortisol at 12 months postpartum. Drawing from the literature that has documented associations between traumatic experiences and disruptions to cortisol dynamics, we hypothesized that traumatic events exert fetal programming effects, and lifetime trauma history would be associated with maternal and infant diurnal cortisol dynamics. As other studies have positive associations between maternal trauma exposure and salivary cortisol concentrations in mothers, infants, and children (Levendosky et al. 2016; Martinez-Torteya et al. 2016), we predicted that maternal exposure to traumatic events would be associated with dysregulated diurnal cortisol, indicated by elevated levels of cortisol at bedtime in the saliva of mothers and infants.

2 | Materials and Methods

2.1 | Study Design

Participants in this preliminary study were part of a larger randomized controlled trial (RCT) of a home-visiting program designed to improve parenting skills and infant development (NCT02807818) for first-time adolescent mothers living in low-income urban regions of São Paulo, Brazil (Alarcão et al. 2021; Fatori et al. 2021). It consisted of a nurse home visit program aimed at promoting mother–infant relationships and child development through biweekly (first and last months of pregnancy and from 2 to 20 months postpartum), weekly (from 2 to 20 months of pregnancy), and monthly (21–24 months postpartum) home visits by trained nurses. Participants randomized to the control group received care as usual through Brazil's universal public health care system. This program is for first-time pregnant adolescents and later their infant, recruited during the first 16 weeks of pregnancy with visits continuing until their child reaches age 2. In brief, 80 adolescents between 14 and 19 years of age and pregnant were recruited in the first trimester, between 8 and 16 weeks of gestation (T1), and completed a baseline interview. Eligibility criteria included living in the western regions of São Paulo and of low socioeconomic status (SES), defined as classes C-D/E, based on the Criteria for Economic Classification (Associação Brasileira de Empresas de Pesquisa 2007), and pregnant for the first time. A second interview (T2) was conducted at 30 weeks of gestation. A third interview (T3) was completed when infants were 12 months old. Pregnancy interviews and sample collection were completed for 62 women. Participants were asked to self-report their age, pre-pregnancy height and weight, and current medication use at the baseline interview. They also self-reported their race and ethnicity using standard Brazilian census categories. Written informed consent was obtained from all the participants and from a parent or guardian years old. The study was approved by the ethical review boards at the University of São Paulo Medical School (ref: 052/15) and the São Paulo Municipal Health Department. Additional approval for analyses of saliva samples was received from the institutional review board at Yale University (2000021354). The analyses presented here were conducted with intervention and control groups combined, with the assigned group included as a dichotomous covariate in the analyses. All infants were without neurological or endocrinological problems.

2.2 | Measures

2.2.1 | Maternal Traumatic Experiences

Maternal exposure to violence and potentially traumatic events was assessed at the postnatal visit (T3) using a checklist-style questionnaire of a locally adapted version of the posttraumatic stress disorder section of the World Health Organization Composite International Diagnostic Interview version 2.1 (CIDI 2.1) (Quintana et al. 2007; Ribeiro et al. 2013; World Health Organization 1997). This version was designed to assess frequently occurring traumatic events in Brazilian urban centers (Ribeiro et al. 2013). The measure includes the original 11 items of the CIDI 2.1 and 20 additional items reflecting traumatic events commonly experienced in Brazilian urban centers, as well as a free-list item, for a total of 31 items. We removed one question about experiences with war as our participants were born in Brazil, and as adolescents, they were not eligible for military service. Participants were also able to free-list events they experienced that were not captured by one of the existing 20 items. Sixty-two participants completed the adapted CIDI questionnaire. Responses to the questionnaire were used to develop seven independent variables representing trauma and violence exposure: (1) a sum score was created to reflect the number of trauma types that participants experienced across their lives. (2) Participants were also asked to retrospectively report if they had experienced any of the events during their pregnancy. The lifetime trauma events were grouped into categories following Ribeiro et al. (2013), including (3) the 10 original CIDI 2.1 items, (4) assaultive violence, (5) other injuries or shocking events, (6) and sudden death or life-threatening injury/illness of a close family member or friend. Finally, participants were asked to report the total number of times they experienced each event. Total lifetime exposure was summed and (7) dichotomized based on a mean split (mean = 3) based on the average in this cohort, following previous work on associations between traumatic events and cortisol in which we investigated differences by average lifetime exposure (Dajani et al. 2018).

2.2.2 | Salivary Cortisol

Maternal and infant saliva samples were collected by mothers at T3. Due to the burdensome nature of the infant saliva collection protocol, we asked mothers to collect only morning and evening samples across 2 days to assess diurnal variation in cortisol production. Infants establish diurnal cortisol rhythms on average at 3–4 months of age (Price, Close, and Fielding 1983; Santiago, Jorge, and Moreira 1996). Mothers were provided four collection tubes for themselves and four infant saliva collection swabs and tubes (Salimetrics, 5004.01-06, 5001.08, 5001.05, State College, PA, United States) and asked to collect saliva samples within 30 min of waking and at bedtime on two consecutive nights. Cortisol outliers with concentrations > 2 SD above the mean were excluded. A total of 40 moms collected morning samples and 37 collected evening samples. After excluding those with incomplete data (those that did not report collection times or were missing one or more samples), the final analyses included 29 moms with morning cortisol samples and 23 with evening samples. The final analyses included 32 infants with complete morning and evening values. Salivary cortisol was quantified

using commercially available enzyme immunoassay (EIA) kits (Arbor Assays, K003-H1, Ann Arbor, MI, United States) at a core laboratory facility at the Institute of Psychiatry at the University of São Paulo Medical School. Samples were run in duplicate, and all samples from each individual and dyad were run on the same plate to reduce variability. Cortisol concentrations were averaged across the two morning and two evening samples to create 1 a.m. and 1 p.m. value for moms and infants. Intra- and inter-assay coefficients of variation were 14.9% and 12.8%, respectively.

2.3 | Statistical Analyses

Summary statistics were generated using mean values and frequencies. Salivary cortisol levels were not normally distributed and were log-transformed (log base 10). Bivariate associations between maternal trauma variables and maternal and infant salivary cortisol levels were conducted using Spearman correlations. We tested for differences in summary trauma variables and covariates between the full sample and the maternal and infant cortisol subsamples using t-tests and binomial difference in proportions tests for interval scale and categorical variables, respectively. We found no statistically significant differences in these variables between full and subsamples (Tables S2–S4). We used multivariable linear regression to investigate associations between trauma variables and maternal and infant salivary cortisol (T3) after controlling for relevant covariates. Covariates in all multivariable analyses included intervention or control group assignment, maternal age, and maternal medication use (other than vitamins or supplements). Infant sex was included in infant models. The relevant morning or evening sample collection time was included in all analyses. Given our small sample size, we used a significance threshold of $p < 0.10$ in bivariate analyses to identify trauma variables to include in multivariable analyses. A threshold of $p < 0.05$ was used for all multivariable regression analyses. All statistical analyses were performed using R version 4.2.2 and SPSS v.24 (IBM). Figures were generated in R using the ggplot2 package (Wickham 2011).

3 | Results

3.1 | Descriptive Data on Trauma Exposure

Characteristics of the participants and summary data about exposure to traumatic events are provided in Table 1. Adolescents were on average 17.1 years old ($SD = 1.26$) at recruitment and self-reported as predominantly mixed race (50%). Summary data on traumatic events are presented in Table 2 for cohort lifetime and pregnancy exposure and exposure during pregnancy for the subsample with complete infant salivary cortisol data. Of the 62 participants that completed the CIDI, 72% reported experiencing at least one of the traumatic event types in their lifetime; on average, adolescents reported 2.45 ($SD = 2.58$) traumatic events over the life course. Half the sample reported experiencing assaultive violence, 47% experienced an accident or injury, and 44% had a family member or friend experienced sudden illness, injury, or death. The most common traumatic experiences reported were the sudden unexpected death of a close family member or friend (26%), followed by witnessing someone being injured or killed (21%), witnessing domestic violence in childhood (21%), being attacked with a weapon

TABLE 1 | Sample characteristics.

Maternal characteristics	
Age, <i>M</i> (<i>SD</i>)	17.1 (1.26)
Pre-pregnancy BMI, kg/m ² , <i>M</i> (<i>SD</i>)	22.1 (3.86)
Race and Ethnicity, <i>n</i> (%), <i>n</i> = 80	
White	25 (31.3)
Black	10 (12.5)
Mixed race	40 (50)
Asian	4 (5.0)
Indigenous	1 (1.3)
CIDI traumatic events (<i>n</i> = 62)	
Traumatic events, <i>M</i> (Median, <i>SD</i>)	2.45 (2.00, 2.58)
Traumatic events in pregnancy, <i>M</i> (Median, <i>SD</i>)	0.52 (0.00, 0.88)
Original CIDI items, <i>M</i> (Median, <i>SD</i>)	1.21 (0.50, 1.57)
Assaultive violence, <i>M</i> (Median, <i>SD</i>)	0.98 (0.50, 1.26)
Injury or accident, <i>M</i> (Median, <i>SD</i>)	0.79 (0.00, 1.12)
Injury, accident, or death of close person, <i>M</i> (Median, <i>SD</i>)	0.66 (0.00, 0.92)
Maternal AM cortisol (<i>n</i> = 29), pg/mL, <i>M</i> (<i>SD</i>)	1666.8 (1441.26)
Maternal PM cortisol, pg/mL (<i>n</i> = 23), <i>M</i> (<i>SD</i>)	1021.1 (781.54)
Infant characteristics	
Male offspring, <i>n</i> (%)	38 (47.5)
Birthweight (g), <i>M</i> (<i>SD</i>)	2982.5 (607.04)
Infant AM cortisol, pg/mL (<i>n</i> = 32), <i>M</i> (<i>SD</i>)	1913.93 (1987.16)
Infant PM cortisol, pg/mL (<i>n</i> = 32), <i>M</i> (<i>SD</i>)	1163.01 (1232.62)

(16%), and seeing or touching a corpse (16%). Roughly 34% of participants who completed the interview recalled experiencing at least one such event during their pregnancy (mean = 0.52, $SD = 0.88$), reported an undisclosed type of shocking event (8%), a shocking event experienced by family or friends (8%), or witnessing someone being killed or injured (6%). The data were similar for both the subsample with complete infant cortisol data, with 34% who reported experiencing at least one traumatic event during pregnancy (mean = 0.56, $SD = 0.91$), and the subsample with complete maternal cortisol data with 35% who reported experiencing at least one traumatic event during pregnancy (mean = 0.61, $SD = 0.99$).

3.2 | Associations Between Trauma and Cortisol

Results of the Spearman correlation were used to identify bivariate associations between maternal variables and log-transformed salivary cortisol values, which are presented

TABLE 2 | Traumatic event frequencies.

Traumatic event	Lifetime exposure, <i>n</i> (%)	Lifetime exposure, maternal cortisol subsample, <i>n</i> (%)	Pregnancy, <i>n</i> (%)	Pregnancy exposure, infant cortisol subsample, <i>n</i> (%)	Pregnancy exposure, maternal cortisol subsample, <i>n</i> (%)
Assaultive violence	31 (50.0%)	10 (43.5%)			
Attacked without a weapon	7 (11.3%)	0	1 (1.6%)	1 (3.1%)	1 (4.3%)
Attacked with a weapon	10 (16.1%)	5 (21.7%)	0	0	0
Kidnapped or held captive	0	0	0	0	0
Fast kidnap	0	0	0	0	0
Torture/terrorism	0	0	0	0	0
Death threats	4 (6.5%)	1 (4.3%)	2 (3.2%)	1 (3.1%)	1 (4.3%)
Conflict between gangs/drug dealers	0	0	0	0	0
Rape	1 (1.6%)	1 (4.3%)	0	0	0
Sexual molestation	3 (4.8%)	1 (4.3%)	0	0	0
Being beaten up by parents/relatives	3 (4.8%)	2 (8.7%)	0	0	0
Being beaten up by an intimate partner	7 (11.3%)	1 (4.3%)	1 (1.6%)	1 (3.1%)	1 (4.3%)
Being beaten up by non-family	5 (8.1%)	3 (13.0%)	1 (1.6%)	1 (3.1%)	1 (4.3%)
Witnessing domestic violence during childhood	13 (21%)	4 (18.4%)	0	0	0
House broken into while at home	2 (3.2%)	1 (4.3%)	1 (1.6%)	1 (3.1%)	1 (4.3%)
Blackmailing telephone calls	7 (11.3%)	0	0	0	0
Other injury or shocking events	29 (46.8%)	14 (60.9%)			
Car/motorcycle accident	6 (7.5%)	2 (8.7%)	3 (4.8%)	1 (3.1%)	2 (8.7%)
Other accidents	1 (1.6%)	0	1 (1.6%)	1 (3.1%)	0
Fire, flood, natural disaster	3 (4.8%)	2 (8.7%)	2 (3.2%)	1 (3.1%)	1 (4.3%)
Witnessing someone being killed or injured	13 (21%)	4 (18.4%)	4 (6.5%)	3 (9.4%)	3 (13.0%)
Witnessing a bank robbery	1 (1.6%)	0	1 (1.6%)	1 (3.1%)	1 (4.3%)
Witnessing a shoot-out or being victim of stray bullet	9 (14.5%)	5 (21.7%)	2 (3.2%)	2 (6.3%)	2 (8.7%)

(Continues)

TABLE 2 | (Continued)

Traumatic event	Lifetime exposure, <i>n</i> (%)	Lifetime exposure, maternal cortisol subsample, <i>n</i> (%)	Pregnancy, <i>n</i> (%)	Pregnancy exposure, infant cortisol subsample, <i>n</i> (%)	Pregnancy exposure, maternal cortisol subsample, <i>n</i> (%)
Having one's house broken into while not at home	5 (8.1%)	2 (8.7%)	0	0	0
Seeing or touching a corpse	10 (16.1%)	5 (21.7%)	1 (1.6%)	0	0
Witnessing atrocities, slaughter, massacre	1 (1.6%)	0	0	0	0
Human-made disaster	0	0	0	0	0
Diagnosis of life-threatening illness	0	0	0	0	0
Sudden death or life-threatening illness of a close person	27 (43.5%)	11 (47.8%)			
Sudden unexpected death of a close person	16 (25.8%)	5 (21.7%)	1 (1.6%)	0	0
Child with life-threatening illness or injury	2 (3.2%)	0	1 (1.6%)	0	0
Shocked by event that happened to someone else	10 (16.1%)	4 (18.4%)	5 (8.1%)	4 (12.5)	2 (8.7%)
Other shocking events	13 (21%)	6 (26.1%)	5 (8.1%)	2 (6.3%)	4 (17.4%)

Note: Trauma interview sample size *n* = 62; infant cortisol subsample size *n* = 32; maternal cortisol subsample *n* = 23.

in Table S1. Maternal morning cortisol levels were not significantly associated with any of the trauma data but were marginally correlated with traumatic events in pregnancy ($r_s = 0.29$, p -value = 0.07). Maternal evening cortisol was significantly associated with lifetime exposure to assaultive violence ($r_s = 0.36$, p -value = 0.03) and marginally correlated with items from the original CIDI 2.1 scale ($r_s = 0.29$, p -value = 0.09). Infant evening cortisol levels were associated with the number of trauma types mothers experienced during pregnancy ($r_s = 0.44$, p -value = 0.01), maternal lifetime exposure to accident or injury ($r_s = 0.37$, p -value = 0.03), and exposure to death, illness, or injury of a family member or close friend ($r_s = 0.38$, p -value = 0.03).

We then adjusted for covariates using multivariable linear regression. Maternal morning cortisol was not significantly associated with traumatic event exposure in pregnancy after adjusting for covariates, including intervention or control group, maternal age, maternal prenatal medication use (other than vitamins or supplements), and saliva collection time (p -value = 0.56). Maternal evening cortisol (Table 3, Figure 1) was higher in mothers with more lifetime exposure to the original CIDI items ($b = 0.16$, p -value = 0.007, Adj. $R^2 = 0.34$, Figure 1A) and with lifetime

exposure to assaultive violence ($b = 0.196$, p -value = 0.020, Adj. $R^2 = 0.41$, Figure 1B). We also examined associations between maternal morning and evening cortisol stratified by the average number of lifetime trauma exposures (≤ 3 traumatic events or > 3 trauma events; Figure 1C). Maternal morning cortisol was not different by lifetime trauma grouping (p -value = 0.98), but maternal evening cortisol was higher in those with ≥ 3 traumatic event types ($b = 0.47$, p -value = 0.028, Adj. $R^2 = 0.39$).

The results of the multivariable regression models predicting infant cortisol levels (Table 4) showed that there was a positive association between the number of trauma events experienced during pregnancy and the evening cortisol levels of infants at 12 months of age ($b = 0.21$, $p = 0.01$, Adj. $R^2 = 0.36$; Figure 1D). However, neither maternal lifetime experience of accident or injury (p -value = 0.07) nor death, injury, or illness of a close friend (p -value = 0.14) were associated with infant cortisol. Infant morning (p -value = 0.66) and evening cortisol (p -value = 0.13) did not differ by mean stratified maternal lifetime trauma groupings (p -value > 0.05). All infant models controlled for included intervention or control group, maternal age, maternal prenatal medication use (other than vitamins or supplements), infant sex, and saliva collection time.

TABLE 3 | Associations between trauma history and maternal salivary evening cortisol 12 months postnatal ($n = 23$).

	<i>B</i>	Standard error	<i>t</i>	<i>p</i>	Confidence interval	
Original CIDI items						
Intervention group	0.244	0.142	1.718	0.111	−0.066	0.554
Maternal age	−0.046	0.045	−1.015	0.330	−0.145	0.053
Medication use	−0.162	0.163	−0.991	0.341	−0.517	0.194
Collection time	−0.033	0.015	−2.146	0.053	−0.066	0.000
Original CIDI items	0.162	0.050	3.230	0.007**	0.053	0.271
Model adj. <i>R</i> ²	0.499					
Assaultive violence						
Intervention group	0.208	0.156	1.330	0.208	−0.133	0.548
Maternal age	−0.044	0.049	−0.890	0.391	−0.152	0.064
Medication use	−0.137	0.176	−0.780	0.450	−0.521	0.246
Collection time	−0.034	0.017	−2.067	0.061	−0.070	0.002
Assaultive violence	0.196	0.074	2.668	0.020*	0.036	0.356
Model adj. <i>R</i> ²	0.412					
Life trauma mean split						
Intervention group	0.206	0.160	1.287	0.222	−0.143	0.555
Maternal age	−0.042	0.051	−0.841	0.417	−0.153	0.068
Medication use	−0.123	0.180	−0.682	0.508	−0.515	0.269
Collection time	−0.035	0.017	−2.102	0.057	−0.072	0.001
Trauma (Ref: trauma ≤ 3)	0.472	0.188	2.508	0.028*	0.062	0.881
Model adj. <i>R</i> ²	0.385					

* p -value ≤ 0.05 .** p -value ≤ 0.01 .

4 | Discussion

This preliminary study examined associations between maternal exposure to violence and other traumatic events and diurnal salivary cortisol in a cohort of first-time adolescent mothers and their infants in São Paulo, Brazil, at 12 months postpartum. We found that adolescent mothers who had experienced traumatic events as indexed by the CIDI 2.1 experienced a higher than the cohort average number of traumatic events, and a greater number of events in the domain of assaultive violence had elevated evening cortisol levels. Infants whose mothers experienced a greater number of traumatic events during pregnancy had higher levels of bedtime salivary cortisol at 12 months of age. These findings are generally consistent with a model of fetal programming (Sandman et al. 2012; Seckl 1998) and the intergenerational transmission of trauma exposure (Bowers and Yehuda 2018; Yehuda and Lehrner 2018) in which maternal experience of violence and trauma may be transmitted intergenerational via infant stress physiology. Lifetime trauma exposure may also become embedded in maternal HPA function in the postpartum period.

We observed a high level of exposure to traumatic events among adolescent mothers in our sample—72% reported experiencing

at least one of the events in their lifetime. Over 1/3 of the participants recalled experiencing at least one event during their pregnancy, suggesting that low-income adolescent mothers may be at a high risk of experiencing traumatic events. Despite the young age of our sample, this is only slightly lower than the lifetime prevalence of 86% reported in a population-representative sample of adults living in São Paulo interviewed from 2007 to 2008 that used the same version of the adapted CIDI inventory (Ribeiro et al. 2013). However, the rates reported by our sample were similar to adolescents enrolled in the 1993 Pelotas Brazilian Birth Cohort Study in which roughly 30% of adolescents experienced a robbery by age 18 (Murray et al. 2018). This prevalence rate is similar to the 61% rate measured in 2010 for an adolescent sample living in São Gonçalo, a city in the metropolitan area outside of Rio de Janeiro characterized as a low-resource and high-crime city (Avanci et al. 2021). Our participants also reported a higher frequency of exposure than a low-income pregnant adolescent population living in the northern region of São Paulo, which reported a 21.9% lifetime prevalence and 2% during pregnancy (Ferri et al. 2007). Such large differences may be partially due to the screening measure as the study by Ferri et al. used the California Perinatal Assessment, which only contains five general items, although it captures similar events.

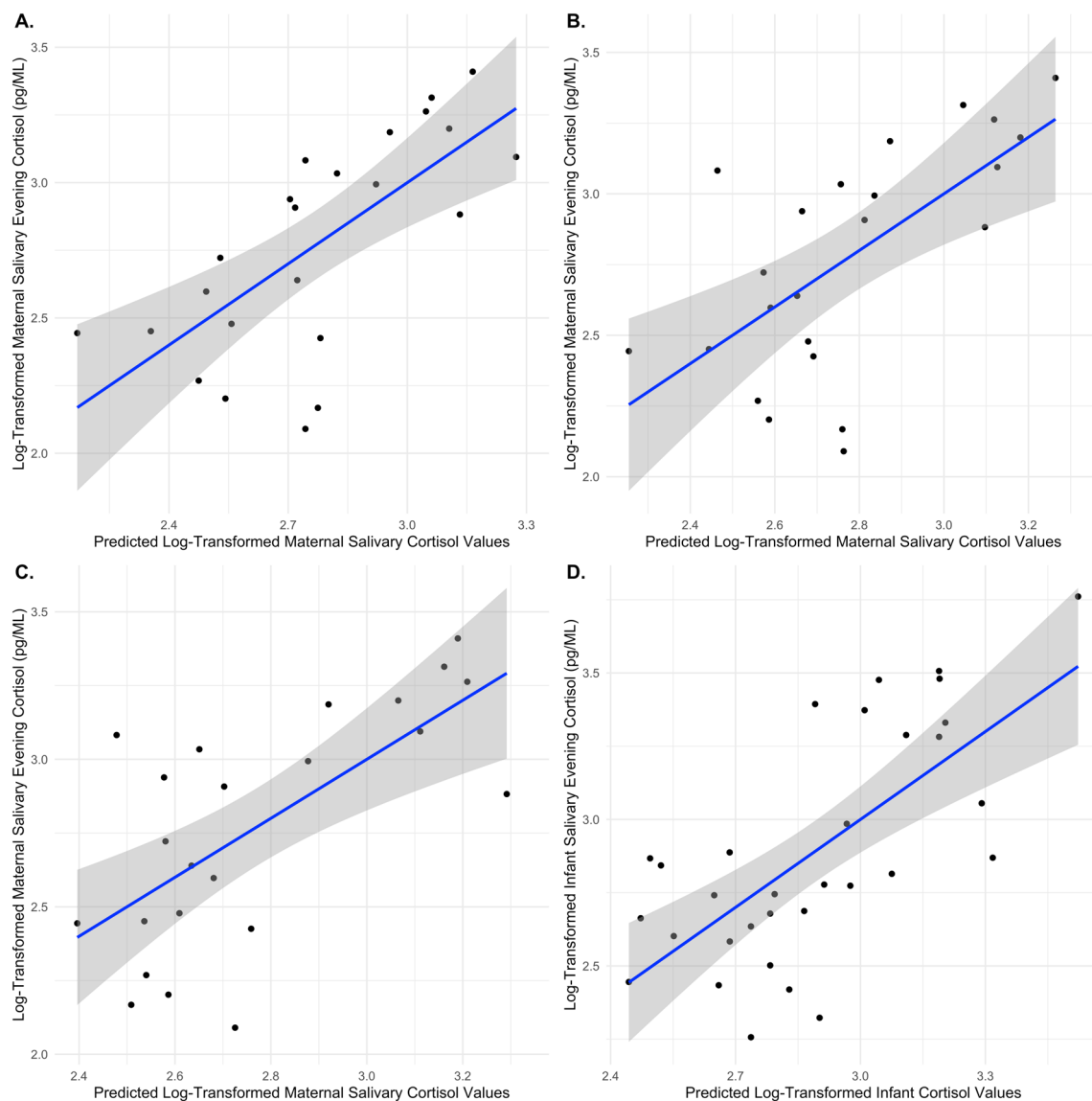


FIGURE 1 | Regression plots of associations between trauma variables and maternal and infant salivary evening cortisol. Regression plots with predicted dependent variables from each multivariable linear regression on the x-axis and the raw dependent variables on the y-axis. Panels (A–C) are multivariable regressions with maternal log-transformed salivary evening cortisol as the dependent variable and (A) the 10 original CIDI items, (B) the assaultive violence items, and (C) a dichotomous variable based on a split stratified by the average number of lifetime trauma exposures (≤ 3 traumatic events or > 3 trauma events) as the independent variables. Panel (D) is the multivariable regression with infant log-transformed salivary evening cortisol as the dependent variable and maternal trauma exposure during pregnancy as the independent variable.

This study also had a greater income range than was reflected in our participants. Taken together, these results are generally in line with the existing work on adolescent motherhood, which suggests that adolescent mothers may be disproportionately vulnerable to violence and trauma relative to their peers or older mothers (Bott et al. 2019). Elevated exposure and subsequent consequences for maternal mental health, physiology, and birth and infant outcomes (Thomas et al. 2019) may partially explain cycles of intergenerational vulnerability (Ferraro et al. 2013).

Lifetime trauma history was associated with mothers' salivary cortisol at 12 months postpartum. Specifically, mothers with higher-than-average lifetime trauma exposure and high exposure to events on the original CIDI 2.1 and in the domain

of assaultive violence had elevated salivary cortisol levels at bedtime. We did not find any significant associations between trauma history and morning cortisol levels. This may not be surprising as elevated evening cortisol levels are suggestive of a pattern of HPA dysregulation that may be predictive of mortality in adulthood (Jones and Gwenin 2021). However, a more robust sampling protocol that assessed the full cortisol awakening response may have detected an association. The lack of association between recalled events experienced during pregnancy is in contrast to previous work that found an association between IPV and afternoon salivary cortisol levels in Nicaraguan women (Valladares et al. 2009). However, this may reflect differences between our more general assessment of trauma exposure and IPV or recall bias.

TABLE 4 | Associations between traumatic events in pregnancy and infant evening cortisol at 12 months postnatal ($n = 32$).

Events in pregnancy	<i>B</i>	Standard error	<i>t</i>	<i>p</i>	Confidence interval	
Intervention group	−0.205	0.115	−1.786	0.086	−0.441	0.031
Maternal age	−0.108	0.041	−2.654	0.014*	−0.192	−0.024
Medication use	0.057	0.141	0.404	0.689	−0.233	0.347
Infant sex	−0.026	0.116	−0.226	0.823	−0.265	0.213
Collection time	−0.003	0.201	−0.016	0.988	−0.417	0.410
Events in pregnancy	0.210	0.064	3.304	0.003**	0.079	0.342
Adj. R^2	0.361					

* p -value ≤ 0.05 .** p -value ≤ 0.01 .

We found that maternal exposure to traumatic events during pregnancy was statistically associated with elevated evening salivary cortisol levels of infants at 12 months of age, suggesting that such exposures may contribute to the fetal programming of infant stress physiology. While this is generally in line with the literature, associations between trauma exposure in specific life stages and offspring cortisol are not consistent across studies. For example, one recent study documented associations between maternal trauma exposure and infant cortisol levels; however, these associations were not significant for events experienced during adolescence or pregnancy (Slopen et al. 2018). A study of 1-month-old infants of Turkish women that reported IPV in pregnancy had higher daytime salivary cortisol levels (Avci et al. 2022). Some variability in results may be due to differences in collection protocols or sample type (hair vs. saliva). While most research has focused on the effects of maternal trauma exposure (specifically IPV) on infant/childhood behavioral outcomes, we contribute to the growing literature investigating the effects on infant HPA function (Hibel et al. 2009; Levendosky et al. 2016; Slopen et al. 2018). Our results suggest that HPA dysfunction may be detectable within the first year of life in infants of mothers with high levels of trauma exposure during pregnancy. Future research is needed to investigate if this early dysregulation continues into childhood and adolescence and to what extent this may or may not shape vulnerability for negative health outcomes later in life.

A significant contribution of this study is the investigation of the intergenerational consequences of maternal trauma exposure on infant HPA function in the context of adolescent motherhood and chronic poverty in a developing country. Existing studies have typically focused on relatively high-income clinical samples and Euro-American communities. While notable exceptions exist (Avci et al. 2022; Bush et al. 2017; Hagaman et al. 2020; Slopen et al. 2018; Valladares et al. 2009), such studies are largely limited to the United States or Europe. One notable exception, a study of the intergenerational impact of maternal IPV in Pakistan, found associations between postnatal, but not prenatal, IPV exposure and levels of an HPA hormone dehydroepiandrosterone, but found no associations with cortisol, in infant hair (Hagaman et al. 2020). Our study extends this research to a low-income and ethnically diverse sample of adolescents exposed to chronic poverty in an urban region of Brazil. Additional research is needed to contribute to understanding

the intergenerational impact of maternal trauma given the conflicting results in this literature.

The mechanisms linking maternal exposure to traumatic events and postnatal HPA function are unclear but may include hormonal or immunological changes during pregnancy. While reviews have linked trauma exposure to dysregulated cortisol patterns for adults (Steudte-Schmiedgen et al. 2016), associations for children and adolescents are unclear and may not become apparent until adulthood (Bunea, Szentágotai-Táttar, and Miu 2017; Gray et al. 2018). Associations between cortisol and psychological distress during pregnancy are inconsistent, and it is unclear how the responsivity of the HPA to stressors may change throughout pregnancy, particularly in low-income populations (Mustonen et al. 2018; Van den Bergh et al. 2005). Other hormones, such as placental corticotrophin-releasing hormone may also be involved and are associated with childhood trauma (Moog et al. 2016; Steine et al. 2020). Normative immunological changes during pregnancy may be disrupted by trauma exposure. For example, experiencing adversity early in life or during pregnancy is associated with higher proinflammatory gene expression phenotypes, and total life adversity is associated with a reduced immunosuppressive gene expression (Aschbacher et al. 2021). Variation in the maternal immune function during pregnancy has been associated with infant outcomes, including within this and other Brazilian cohorts (Euclydes et al. 2024). Trauma-induced changes in DNA methylation may also explain how maternal trauma influences offspring cortisol levels (Hjort et al. 2021). Future studies are needed to examine how such changes in pregnancy may relate to postnatal outcomes.

There may also be potential postnatal pathways that contribute to infant developmental outcomes or reinforce fetal programming pathways. Factors such as early life adversity, lifetime trauma exposure, and PTSD symptoms have been associated with higher levels of parenting stress or negative parenting practices relative to unexposed individuals (Christie et al. 2019; Lotto, Altafim, and Linhares 2021). Parenting stress, or adverse psychological stress to the demands of parenting (Deater-Deckard 1998), may impact infant and child outcomes by influencing parental perceptions of parenting efficacy parenting practices, or parent–child interactions (Carapito et al. 2020; Mak et al. 2020; Moreland et al. 2016). Such processes may also be impacted during the prenatal period as maternal–fetal

attachment is associated with postnatal parenting stress (Hruschak et al. 2022). These prenatal and life course factors may also disrupt pre- and postnatal attachment and bonding processes that may contribute to adverse outcomes for infants (Bosquet Enlow et al. 2014; Erickson, Julian, and Muzik 2019; Kjerulff et al. 2021). However, these postnatal process studies identify parenting stress, efficacy, and parent–infant attachment as opportunities for intervention to buffer the impact of these prenatal programming processes (Erickson, Julian, and Muzik 2019). Resilience to trauma may also modify the effects of maternal trauma and PTSD attachment and bonding (Nie, Pan, and Liu 2023). While we did not detect a statistically significant association of the intervention with infant cortisol levels, previous work by our group identified positive impacts of this intervention on infant language development and secure attachment, as well as maternal emotional and verbal responsivity to her infant (Alarcão et al. 2021; Fatori et al. 2021). Larger future studies are needed to examine how such postnatal factors may reinforce or modify the impact of prenatal trauma exposure and PTSD on infant HPAA development and efficacious interventions that provide opportunities to intervene in this process.

The results of this study should be considered with respect to several limitations. First, the analysis is based on a modest sample size. We used cortisol data for only those who had completed the full collection protocol, which reduced the sample size and may induce a selection bias. Our collection protocol may have been too burdensome for the mothers participating in our study, given their economic and time constraints. Emerging techniques for hair or fingernail cortisol sampling may reduce the burden on participants and improve protocol compliance (Phillips et al. 2021; Russell et al. 2012). There were no differences in trauma exposure between those who completed and did not complete the collection protocol. Null findings may be due to a lack of statistical power. Second, we cannot rule out possible genetic influences or liabilities shaping the susceptibility of the HPAA to dysregulation by trauma exposure. Third, the trauma recall measure was retrospective and administered at the 12-month postnatal interview and may be susceptible to recall bias. We attempted to reduce recall bias by using a standardized measure and only asking whether or not the event had occurred, excluding the collection of any subjective details, such as perceived severity. This method has been shown to reduce recall bias in other samples (Hardt and Rutter 2004), and recall bias is shown to be less of a concern for traumatic events than other stressful events (Lalande and Bonanno 2011). The trauma assessment did not include self-assessments of severity, which may modify associations between trauma exposures and health or physiological outcomes. Fourth, the assay performance for cortisol quantification was suboptimal as indicated by the relatively high assay CVs, and future studies are needed to replicate our results. Additionally, we collected only maternal trauma history, and we did not assess PTSD symptoms, paternal trauma exposure, exposure specifically during the postpartum, or other potentially traumatic events experienced by the infants. Paternal exposures during the preconception period and infant postnatal experiences may also influence developmental outcomes (Challacombe et al. 2023), and future studies should work to include fathers in research. Finally, future studies could consider additional potentially relevant covariates, such as infant

gestational age at birth and birthweight, breastfeeding status, maternal history of disease, and indicators of socioeconomic position, such as household assets.

5 | Conclusions

This preliminary study, utilizing a cohort of adolescent mothers and their infants in São Paulo, Brazil, found that maternal trauma exposure was associated with dysregulation of the hypothalamic–adrenal–pituitary axis physiology of mothers and their infants at 12 months postpartum. These findings contribute to a growing literature about the intergenerational consequences of trauma exposure. Future work with larger sample sizes is needed to replicate our results, to consider other biological pathways in the intergenerational transmission of maternal trauma, including the immune system, and to identify potential postpartum modifying factors. Future studies should also consider other aspects of infant health and development, such as cognitive development, temperament, and risk for internalizing or externalizing problems. Future studies should also investigate how dysregulated infant stress physiology is related to risk for adverse developmental and health outcomes and may serve as a mechanism linking early life experience to later adverse outcomes, as well as interventions that may buffer this risk.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.